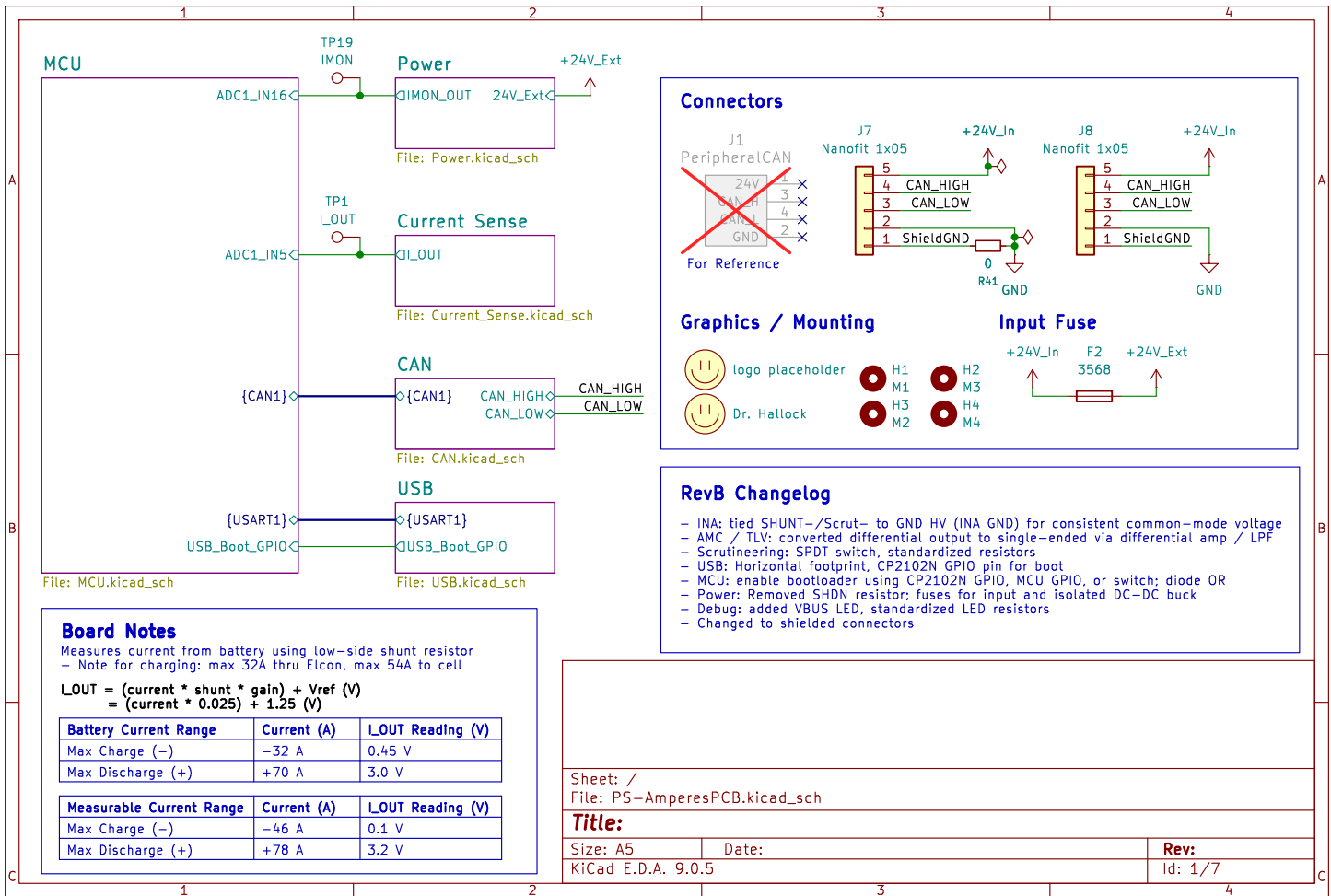


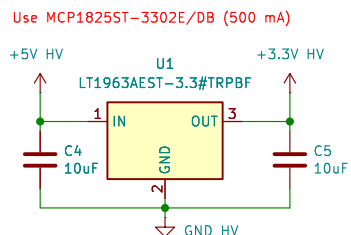
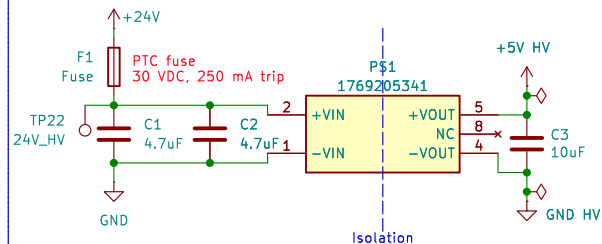


# Power

## 3.3V HV (Isolated)

24V → 5V Buck

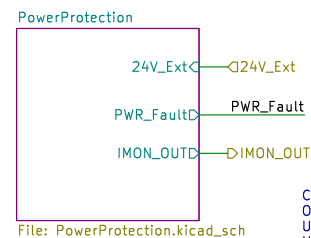
5V → 3.3V LDO



Max Output Current: 0.2A

## Power Protection

Taken from PSOM

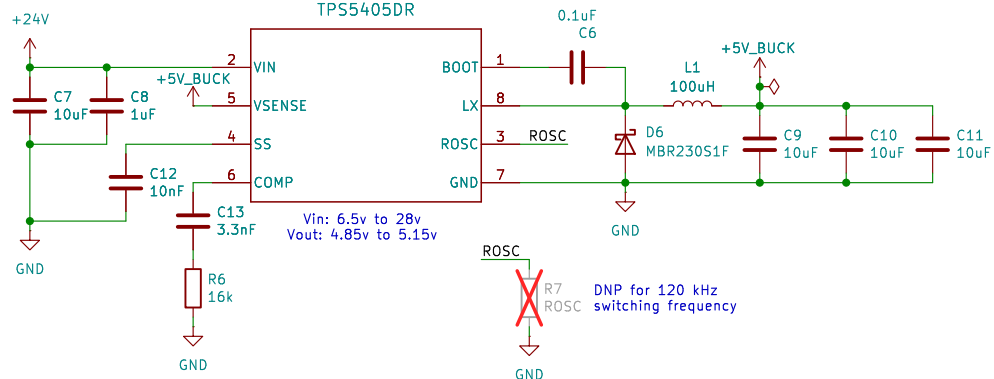


Current Limit: 1.2A  
Overvoltage: ~27v  
Undervoltage: ~20v  
Use 1% resistors  
for closest values

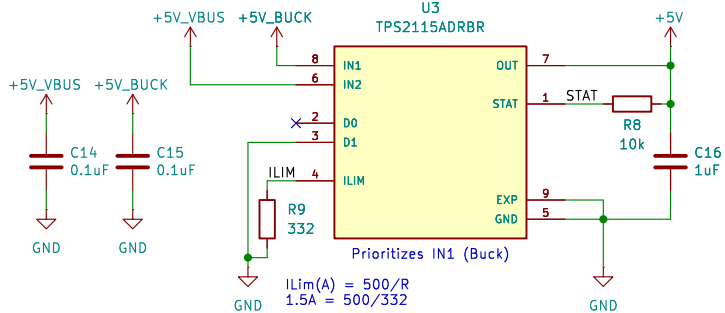
## 3.3V LV (MCU, CAN, etc)

Taken from PSOM

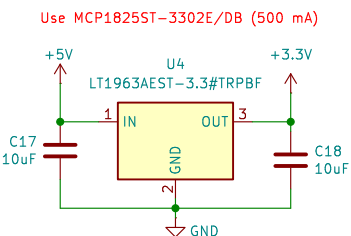
24V → 5V Buck



5v Power Mux



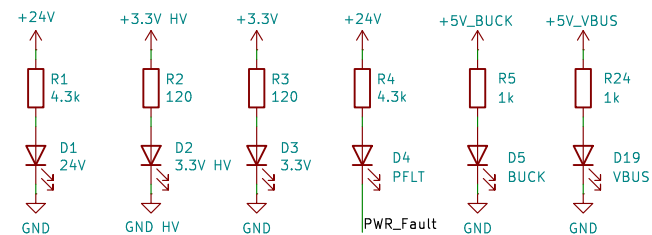
5V → 3.3V LDO



Max Output Current: 1.5A

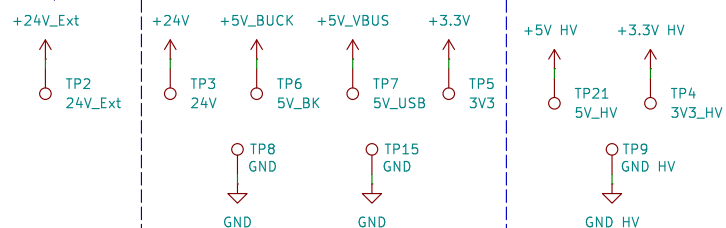
## Indicators

Fault active low



## Testpoints

LV



Sheet: /Power/  
File: Power.kicad\_sch

Title:

Size: A4

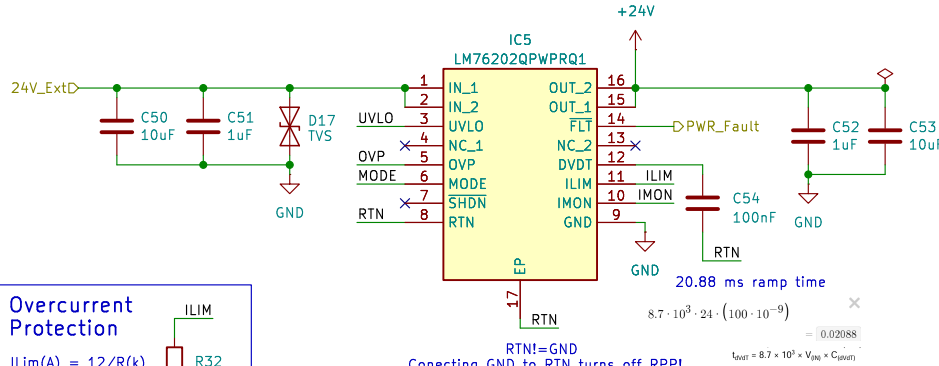
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Rev:

Id: 2/7

Taken from PSOM



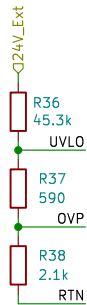
### Overcurrent Protection

$$I_{Lim}(A) = 12/R(k)$$

$$1.2A = 12/10k$$



### Under/Overvoltage Protection



|      | Setpoint | Ideal | Actual |
|------|----------|-------|--------|
| UVLO | 20v      | 1.1v  | 1.12v  |
| OVP  | 27v      | 1.1v  | 1.18v  |

OVP set to 27v for buck max voltage of 28v

$$\left( \frac{2.1 + 0.59}{2.1 + 0.59 + 45.3} \right) 20$$

$$= 1.1210668894$$

$$\left( \frac{2.1}{2.1 + 0.59 + 45.3} \right) 27$$

$$= 1.18149614503$$

### Functional Mode Selection

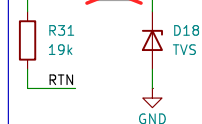
Set to circuit breaker (latchoff)

Use 0 ohm resistor for auto-retry (540ms)



### Current Monitoring

R30 DNP until validated



Typical gain: 78.28 uA/A  
Vout @ 1.3A = 2.407v  
R(imon) = 23.7 mΩ

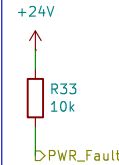
$$R_{(IMONmax)} = \frac{\min[(V_{in} - 1.5), 4 V]}{1.8 \times I_{(LIM)} \times GAIN(IMON)}$$

$$\text{For } I_{(LIM)} > 50 \text{ mA, } V_{(IMON)} = [I_{(OUT)} \times GAIN(IMON)] \times R_{(IMON)}$$

$$\text{For } I_{(LIM)} < 50 \text{ mA (typical), IMON output current is close to } I_{(MAX,OUT)}$$

$$\text{With } R_{(IMON)} = (R_{(MAX,OUT)} - R_{(IMON)})$$

### Pullups



RIP shdn resistor

Sheet: /Power/PowerProtection/  
File: PowerProtection.kicad\_sch

Title:

Size: A5

Date:

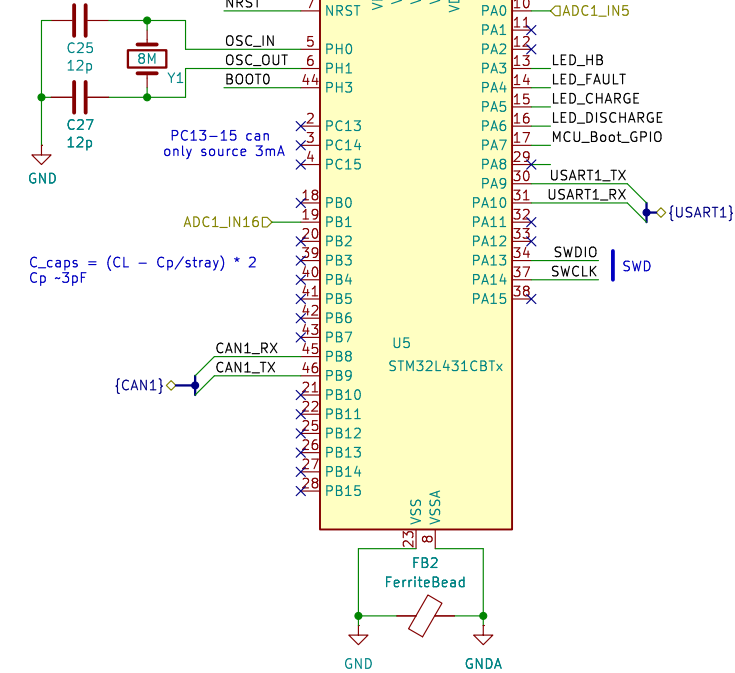
KiCad E.D.A. 9.0.5

Rev:

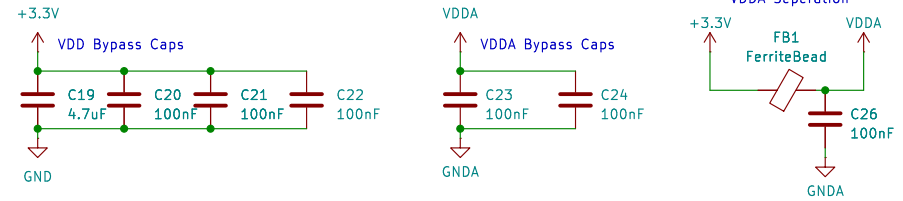
Id: 7/7

## MCU

The crystal and caps have to be as close to the uC as possible and no traces should go underneath these components or connections.

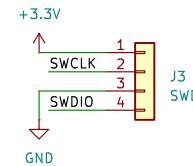


## Bypass, Separation

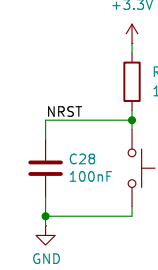


## SWD Connector

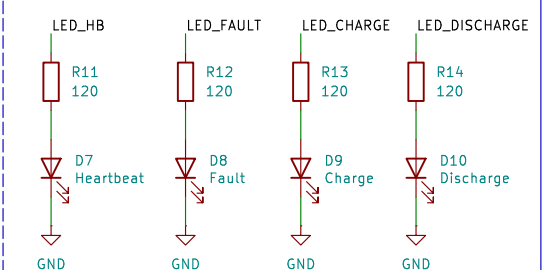
Moxie Nanofit 1053091104  
Mates with 0022013047



## Reset Button



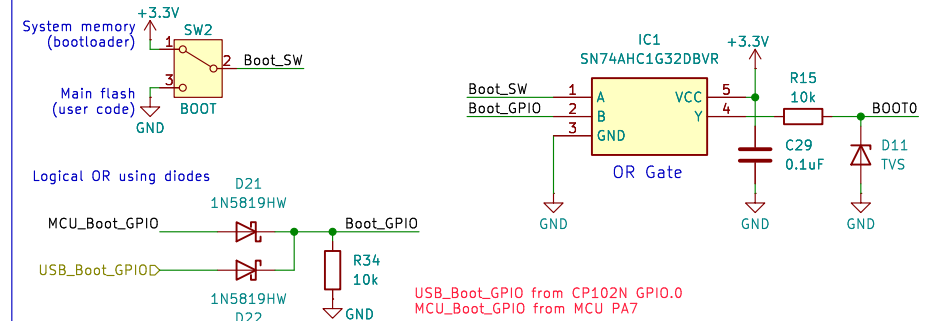
## LED Indicators



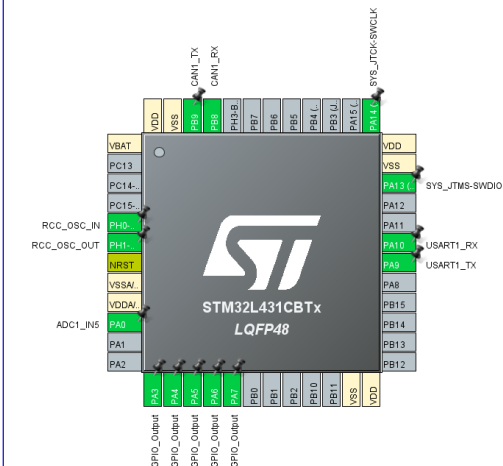
## USART bootloader pins (same as PSOM)

|               |        |                                                                   |
|---------------|--------|-------------------------------------------------------------------|
| USART1_RX pin | Input  | PA10 pin: USART1 in reception mode. Used in input no pull mode.   |
| USART1_TX pin | Output | PA9 pin: USART1 in transmission mode. Used in input no pull mode. |

## Boot Selector



## Pin Functions



Sheet: /MCU/  
File: MCU.kicad\_sch

## Title:

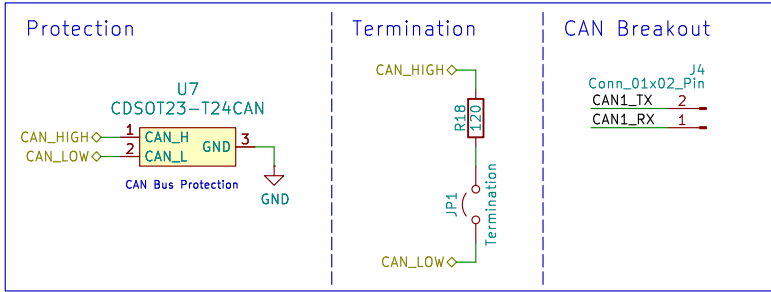
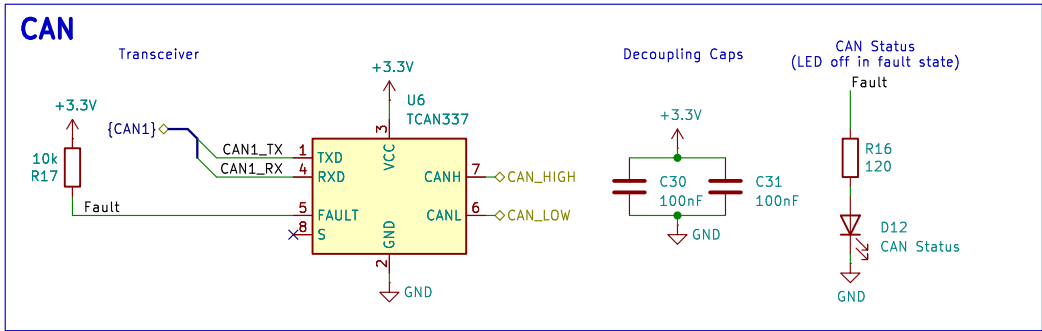
Size: A4

Date:

Rev:

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Id: 3/7



|                                     |       |         |
|-------------------------------------|-------|---------|
| Sheet: /CAN/<br>File: CAN.kicad_sch |       |         |
| <b>Title:</b>                       |       |         |
| Size: A5                            | Date: | Rev:    |
| KiCad E.D.A. 9.0.5                  |       | Id: 4/7 |

## Horizontal USB C Port

Horizontal USB C Port

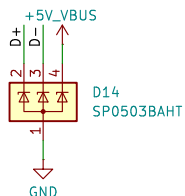


Diagram illustrating the connection of the CP2102N-A02-GQFN24 module to the MCU and power sources.

**Power Connections:**

- VDD Bypass:** +3.3V connected to C32 (4.7uF) and C33 (0.1uF), which are connected to GND.
- VREGIN Bypass:** +3.3V connected to C35 (4.7uF) and C36 (0.1uF), which are connected to GND.
- VBUS:** +5V\_VBUS connected to R20 (22k) and R21 (47k), which are connected to GND. The voltage divider is referenced to the datasheet (pg. 9).

**Signal Connections:**

- UART:** UART\_TX is connected to MCU\_RX. UART\_RX is connected to MCU\_TX.
- USART1:** USART1\_RX is connected to TXD. USART1\_TX is connected to RXD.
- GPIO:** USB\_Boot\_GPIO is connected to TXT/GPIO.0.

**Module Pinout (U8 CP2102N-A02-GQFN24):**

- Pin 1: GND
- Pin 2: GND
- Pin 3: D+
- Pin 4: D-
- Pin 5: VIO
- Pin 6: VDD
- Pin 7: VREGIN
- Pin 8: VBUS
- Pin 9: RST
- Pin 10: TXD
- Pin 11: RXD
- Pin 12: RTS
- Pin 13: CTS
- Pin 14: DSR
- Pin 15: DTR
- Pin 16: DCD
- Pin 17: RI/CLK
- Pin 18: SUSPEND
- Pin 19: SUSPEND
- Pin 20: TXT/GPIO.0
- Pin 21: RXT/GPIO.1
- Pin 22: RS485/GPIO.2
- Pin 23: WAKEUP/GPIO.3
- Pin 24: GND

**Title:**

|                    |  |
|--------------------|--|
| Size: A5           |  |
| KiCad E.D.A. 9.0.5 |  |

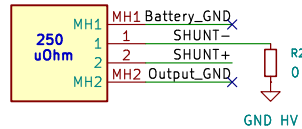
Date:

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Id: 5/7

## Input

### 250uOhm Shunt Resistor

R39  
WSBS8518L2500JTP3

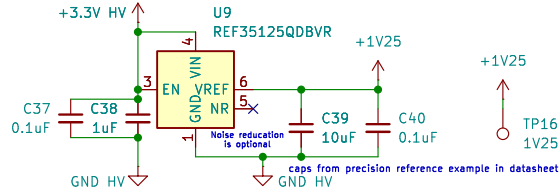


Footprint is M4 version; edit it to have M3 holes

SHUNT- SHUNT- TP11  
SHUNT+ SHUNT+ TP10

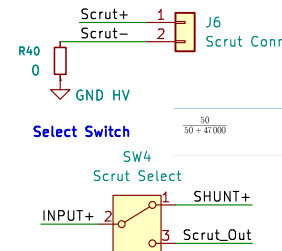
Expected current range: -32 to 70 A  
w/ corr. voltage range: -8 to 17.5 mV  
Low-side shunt, so current path is as follows:  
Discharge: Output\_GND -> Battery\_GND  
Charge: Battery\_GND -> Output\_GND

### Voltage Reference (1.25V)

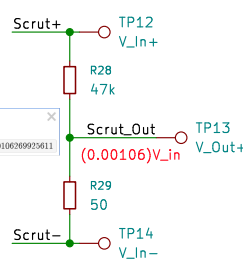


## Current Scrutineering

### Input



### Divider



### DIRECTIONS

- Pass in voltage from scrutineering board
- Use switch to select between shunt / scrutineering inputs

### RELATIONSHIP

- \*  $L_{OUT} = ((0.00106)V_{in} * 100) + 1.25 \text{ (V)}$
- \* Spoof Current =  $(0.00106)V_{in} / (250 \text{ uOhm}) \text{ (A)}$

### Example

| Input (V) | Scrut_Out (mV) | Spoof Current (A) | L_OUT Reading (V) |
|-----------|----------------|-------------------|-------------------|
| -10 V     | -10.6 mV       | -42.5 A           | 0.19 V            |
| +17 V     | +18 mV         | +72.2 A           | 3.05 V            |

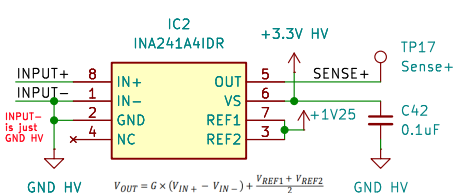
1% tolerance on resistors

SHUNT- and Scrut- connected to GND HV to prevent common-mode voltage from being out of INA's range

## Current Sense Path

Shunt -> current sense amp -> isolation -> filter

### Bidirectional current sense amp



Gain: 100 V/V (INA241 A4 version)

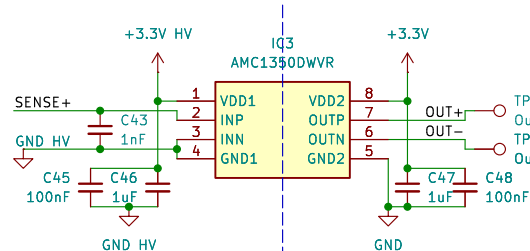
Reference voltage (bias for discharge): 1.25V

- $(70A * 250 \text{ uOhm} * 100V) + V_{ref} = 3 \text{ V}$
- $(-32A * 250 \text{ uOhm} * 100V) + V_{ref} = 0.45 \text{ V}$

Maximum current range: -46 to +78A (0.1 to 3.2V output)

Make sure common-mode stuff is within range

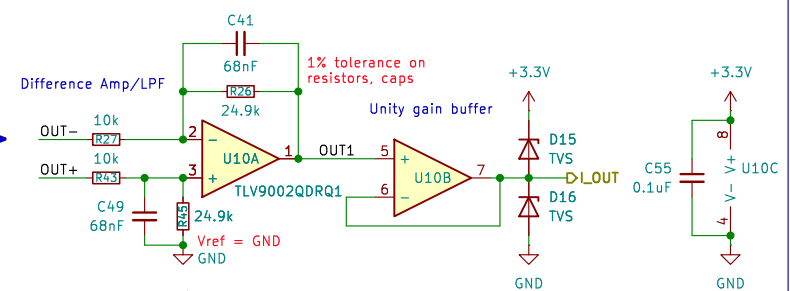
### Isolation



Fixed Gain: 0.4 V/V  
Target Voltage Ranges:  
\* Input: 0.1 - 3.2 V  
\* Output: 0.04 - 1.28 V

OUTN should NOT be tied to GND

### Amplification + Low-Pass Filter



Differential Amplifier / Active LPF  
\*  $V_{out} = [(OUT+) - (OUT-)] * Gain$   
\* Gain = 2.49 V/V  
\* Cutoff freq = ~94 Hz  
\* 1st order lowpass: -20 db/dec rolloff

See LPF documentation in Amperes Design folder

Target Voltage Ranges:  
\* Input: 0.04 - 1.28 V  
\* Output: 0.01 - 3.19 V

Table 8-1. R\_SENSE Selection and Power Dissipation<sup>(1)</sup>

| PARAMETER   | EQUATION                                 | RESULTS AT $V_{IN} = 5V$            |                |                |                |                |
|-------------|------------------------------------------|-------------------------------------|----------------|----------------|----------------|----------------|
|             |                                          | A1, B1 DEVICES                      | A2, B2 DEVICES | A3, B3 DEVICES | A4, B4 DEVICES | A5, B5 DEVICES |
| G           | Gain                                     | 100V/V                              | 200V/V         | 500V/V         | 1000V/V        | 2000V/V        |
| $V_{SENSE}$ | Ideal differential input voltage         | $V_{SENSE} = V_{OUT} / G$           | 500mV          | 250mV          | 100mV          | 50mV           |
| $R_{SENSE}$ | Current sense resistor value             | $R_{SENSE} = V_{SENSE} / I_{MAX}$   | 50mOhm         | 25mOhm         | 10mOhm         | 5mOhm          |
| $P_{SENSE}$ | Current-sense resistor power dissipation | $P_{SENSE} = I_{MAX}^2 * R_{SENSE}$ | 0W             | 2.5W           | 1W             | 0.5W           |

Gain: 100 V/V  
Rsense = 250 uOhm  
Psense = Rsense \* (70A)^2 = 1.225 W

Picked shunt first, then adjusted gain to meet ADC range  
If additional gain is needed, tweak filter circuit

## Current to L\_OUT Conversion

$$L_{OUT} = (\text{current} * \text{shunt} * \text{gain}) + V_{ref} \text{ (V)}$$

## ADC to Current Conversion

$$\text{Current} = ((ADC * (3.3 / 4096)) - V_{ref}) / (\text{Shunt} * \text{gain}) \text{ (A)}$$

Vref = 1.25 V | Shunt = 250 uOhm | Gain = -100 V/V (100 \* 0.4 \* 2.49)  
Shunt \* gain = 250 uOhm \* 100 V/V = 0.025

Sheet: /Current Sense/  
File: Current\_Sense.kicad\_sch

### Title:

Size: A4 Date:

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Id: 6/7